

TTS: Text-to-Speech Systems

From Tacotron 2 to VALL-E — neural TTS architectures, the vocoder problem, and zero-shot voice cloning

9 min

Duration

B2 – Understand

Bloom's

L2 – Intermediate

Level

TH-000008

Prerequisite

Session Overview

What you will learn and the evolution of neural speech synthesis

What You Will Learn

- Why mapping text to natural speech is harder than it looks
- Generation 1: Tacotron 2 + WaveNet (2018)
- Generation 2: FastSpeech 2 + HiFi-GAN (2020–21)
- Generation 3: VALL-E — TTS as language modelling (2023)
- The vocoder problem: Mel spectrogram → waveform
- Zero-shot voice cloning: 3 seconds to any voice
- Ethics and responsible deployment of TTS systems

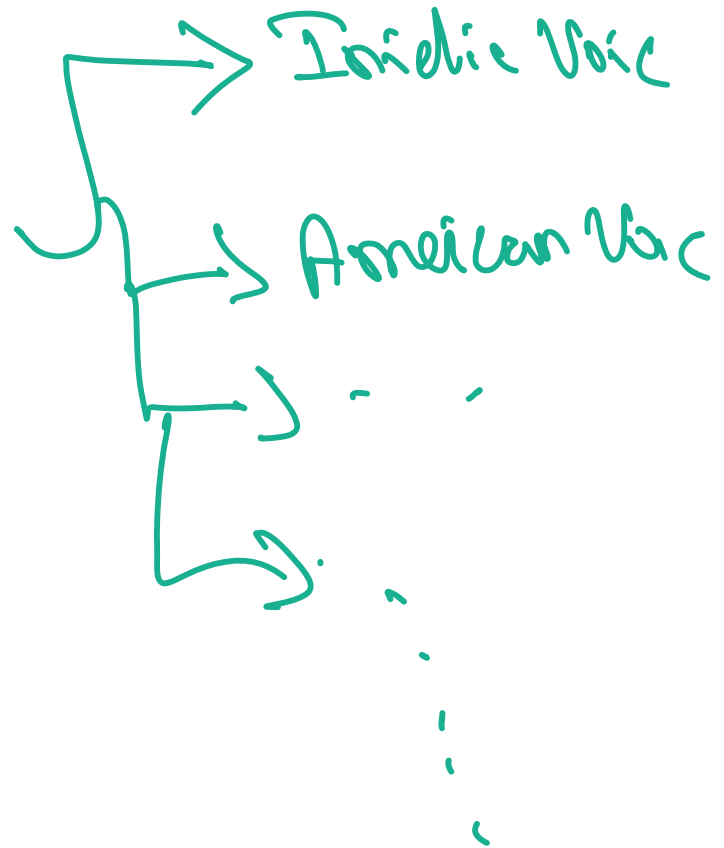
Concept at a Glance

- Duration: 9 minutes
- Bloom's Level: B2 – Understand
- Difficulty: L2 – Intermediate
- Module: 4.2 Audio and Speech Models
- Prerequisite: ASR Models (TH-000008)
- Next: Multimodal Fusion Strategies (TH-000010)

2. Task (STT)

TTS:

Text: "Hi How are you"



Text
one



Not a function at all

Learning Objectives

By the end of this session you will be able to:

9 min

Duration

B2 – Understand

Bloom's

L2 – Intermediate

Level

TH-000008

Prerequisite

- 1 Explain the core challenges that make text-to-speech synthesis difficult
- 2 Describe the Tacotron 2 + WaveNet architecture and identify why it was replaced
- 3 Trace the FastSpeech 2 forward pass with explicit duration, pitch, and energy predictors
- 4 Explain how VALL-E uses a language model over EnCodec tokens for zero-shot voice cloning
- 5 Identify the vocoder problem and explain how HiFi-GAN solves it with a GAN architecture
- 6 Recognise the ethical implications of zero-shot voice cloning systems

SECTION 1 — WHY TTS IS HARD

The Core Challenges of Speech Synthesis

Text contains far less information than speech — prosody, pronunciation, emotion must be inferred

Four Fundamental Challenges in TTS

Mapping characters to speech requires resolving ambiguities that do not exist in written text

The Homograph Problem

'Read' — past tense /rɛd/ or present /ri:d/? '2024' — 'two thousand twenty-four' or 'twenty twenty-four'? Wound — injury /wu:nd/ or wrapped /waʊnd/? Context determines pronunciation; text rarely makes it explicit.

The Prosody Problem

'I didn't say he stole the money' has 7 distinct meanings depending on which word receives primary stress. Text carries zero prosodic information. TTS must infer stress, pacing, and intonation from context alone.

The Naturalness Gap

Humans detect synthetic speech in <200 ms. Early TTS (concatenative, formant synthesis) sounded robotic. Neural TTS narrows this gap dramatically — ElevenLabs achieves MOS 4.6/5 — but has not fully closed it.

The Vocoder Problem

Speech synthesis generates a Mel spectrogram (continuous, compact). Phase information was discarded by the STFT. Inverting a Mel spectrogram back to a waveform requires learning to recover phase — the vocoder problem.

Market scale: ElevenLabs (\$1.1B valuation 2024), Microsoft Azure TTS, Google Cloud TTS, Amazon Polly — commercial TTS generates billions in revenue. The race is driven by naturalness (MOS score) and voice cloning capability.

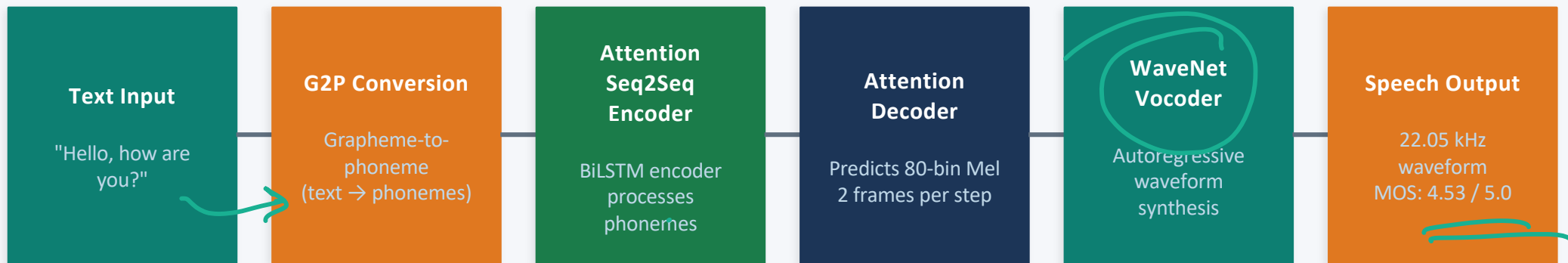
SECTION 2 — GENERATION 1: TACOTRON 2

Tacotron 2 + WaveNet

The first neural TTS to achieve near-human naturalness — and why it was too slow for production

Tacotron 2 Architecture: Text to Mel to Waveform

Two-stage pipeline: attention seq2seq predicts Mel spectrogram; WaveNet vocoder converts to audio



The bottleneck: WaveNet is autoregressive: generates one audio sample at a time. At 22,050 samples/second, synthesising 1 second of speech requires 22,050 sequential neural network forward passes. This makes WaveNet ~1,000× slower than real-time on CPU — unusable for production.

Tacotron 2: The Attention Mechanism

The key innovation: learning to align text and speech automatically — no forced alignment required

Why Attention Is the Core Innovation

The decoder generates one Mel frame at a time and must 'look at' the right part of the phoneme sequence. Tacotron 2 learns this alignment automatically via content-based attention — no duration labels, no forced alignment algorithm, no pronunciation dictionary needed. The model discovers that 'h' = ~3 frames, 'ee' = ~8 frames, etc. from audio alone.

Encoder:	2-layer BiLSTM encodes phoneme sequence into context vectors. Learns phoneme-level acoustic context.
Location-Sensitive Attention:	Attends to encoder outputs at each decoder step. Accumulates attention to ensure monotonic progression through phonemes — doesn't skip or repeat.
Decoder:	2-layer LSTM autoregressive decoder. Predicts 2 Mel frames per step with a pre-net and stop token prediction.
Post-Net:	5-layer 1D CNN applied after decoder. Refines the Mel spectrogram prediction — adds fine spectral detail that the LSTM decoder misses.

Google impact: Tacotron 2 (2018) became Google Assistant's voice for Android and Google Home devices. MOS of 4.53 vs 4.58 for human speech — the first time neural TTS was indistinguishable from humans on clean recordings.

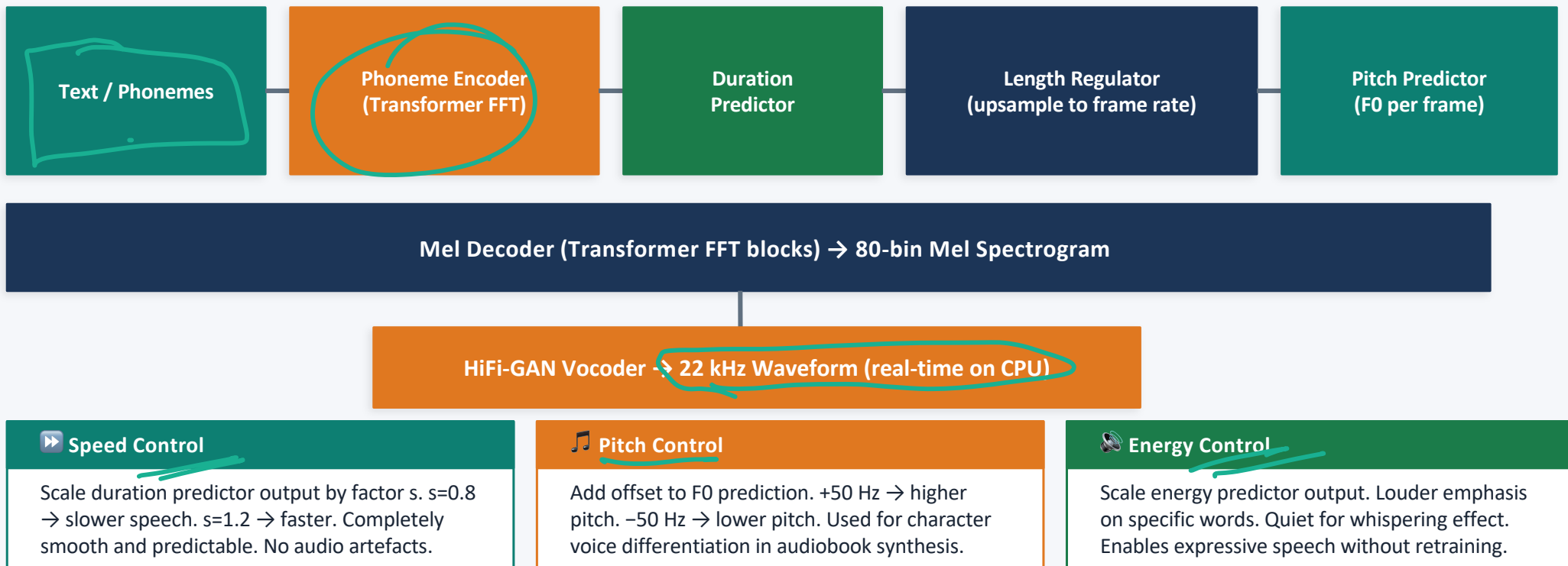
SECTION 3 — GENERATION 2: FASTSPEECH 2

FastSpeech 2 + HiFi-GAN

Parallel synthesis with explicit prosody control — 100× faster than Tacotron 2

FastSpeech 2: Non-Autoregressive Parallel Synthesis

Ren et al. (Microsoft, 2020) — predict all Mel frames simultaneously with explicit duration, pitch, and energy



Speed comparison: Tacotron 2 + WaveNet: $\sim 0.001\times$ real-time (CPU). FastSpeech 2 + HiFi-GAN: $\sim 10\times$ real-time (CPU), $500\times$ real-time (GPU). This is what made neural TTS deployable at scale: Microsoft Azure TTS, Edge read-aloud, NVIDIA Riva.

HiFi-GAN: The Modern Vocoder

Kong et al. (2020) — generative adversarial network converts Mel spectrograms to speech-quality waveforms in real-time

The Vocoder Problem: Recovering Phase from Magnitude

The STFT discards phase information. A Mel spectrogram only has magnitude (energy per time-freq bin). To convert back to a waveform, you must either (1) use Griffin-Lim iterative phase estimation [slow, poor quality] or (2) learn a neural vocoder that recovers phase implicitly. HiFi-GAN chooses (2).

Generator

Stack of transposed convolution layers upsamples Mel ($80 \times T$) to waveform ($22050 \times T$). Multi-receptive field fusion: parallel 1D dilated convolutions at different kernel sizes capture local and global dependencies.

Multi-Period Discriminator (MPD)

5 sub-discriminators process waveform at periods $\{2, 3, 5, 7, 11\}$. Each sees the signal from a different temporal viewpoint. Ensures periodic structure (pitch) is correct.

Multi-Scale Discriminator (MSD)

3 sub-discriminators process raw waveform at different resolutions ($1\times$, $2\times$, $4\times$). Evaluates the signal at multiple temporal scales simultaneously — catches both fine and coarse quality issues.

Performance: HiFi-GAN: MOS 4.4/5.0. Inference: $167\times$ real-time on GPU, $13.4\times$ real-time on CPU. File size: $\sim 14\text{M}$ parameters. The standard open-source vocoder 2021–present. Used with FastSpeech 2, Glow-TTS, JETS, and most modern TTS systems.

SECTION 4 — GENERATION 3: VALL-E

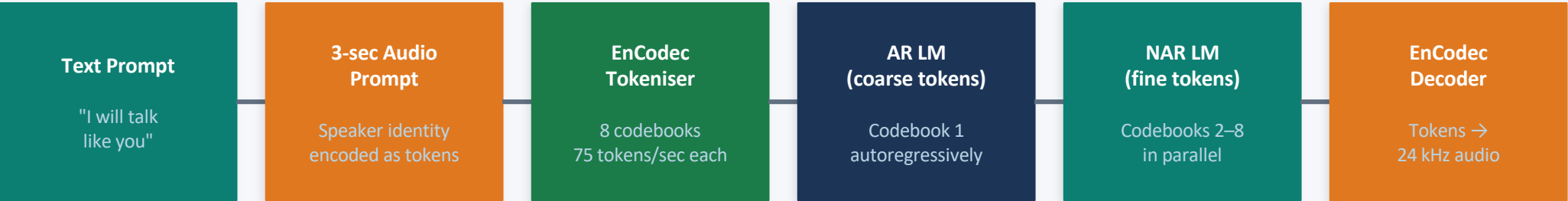
VALL-E: TTS as Language Modelling

Zero-shot voice cloning from 3 seconds — the paradigm shift from Mel spectrograms to discrete audio tokens

VALL-E: Neural Codec Language Model

Wang et al. (Microsoft, 2023) — treat TTS as a conditional language modelling problem over discrete audio tokens

Core idea: Replace continuous Mel spectrograms with discrete audio tokens from EnCodec (Meta's neural audio codec). Train a language model to predict codec tokens conditioned on a 3-second speaker audio prompt. The same LLM training recipe that works for text now works for speech.



Property	Traditional FastSpeech 2	VALL-E
Training data required	Hours of studio recordings	3 seconds of any audio
Speaker adaptation	Full re-recording in TTS voice	Zero-shot from audio prompt
Intermediate representation	Mel spectrogram (continuous)	EnCodec tokens (discrete)
Model architecture	Seq2seq / parallel Transformer	Autoregressive LLM
Emotional/style transfer	Explicit conditioning required	Implicit from audio prompt

⚠️ **Ethics:** Zero-shot voice cloning enables deepfakes. Production systems MUST require explicit speaker consent. Audio watermarking (AudioSeal, Meta) is becoming mandatory. Never deploy voice cloning without consent verification.

SECTION 5 — EVALUATION AND APPLICATIONS

TTS Evaluation and Real-World Deployment

How to measure TTS quality and where these systems are used in production

TTS System Comparison and Deployment Guide

Choose your TTS system based on naturalness, speed, voice flexibility, and deployment constraints

System	Architecture	MOS	Speed	Voice Flexibility
Tacotron 2 + WaveNet	Seq2seq + autoregressive	4.53	0.001× RT (CPU)	Single speaker per model
FastSpeech 2 + HiFi-GAN	Parallel Transformer + GAN	4.4	10× RT (CPU)	Multi-speaker with embedding
VITS	End-to-end flow-based	4.43	Near real-time	Multi-speaker, zero-shot w/ SV
VALL-E (token LM)	LLM + EnCodec codec	4.7	Slower (AR)	Zero-shot from 3 sec prompt
ElevenLabs (commercial)	Undisclosed neural	4.6+	Real-time API	Zero-shot cloning, emotional
Microsoft Azure TTS	FastSpeech 2 variant	4.5	Real-time	Neural voices + custom voice

MOS (Mean Opinion Score): Rated 1–5 by human listeners. 4.0+ is considered very good. 4.5+ is near-human. Human speech MOS on a TTS evaluation set is typically 4.58. MOS is subjective and language/accent dependent — always evaluate on your target population.

TTS in Production: GenAI Applications

Text-to-speech is now a core component of the multimodal AI stack

Production Systems Using Neural TTS

Google Assistant Tacotron 2 neural voice (2018). WaveRNN later replaced WaveNet for speed. 8 neural voices per language.

Microsoft Edge FastSpeech 2 read-aloud for full documents in real-time. Deployed in 40+ languages.

Apple Siri (iOS 17+) On-device neural TTS with sub-100 ms latency. Privacy-preserving — no audio leaves device.

ElevenLabs VALL-E style zero-shot cloning. Emotional speech. Used by audiobook producers, content creators.

Ethical Considerations and Safeguards

Consent verification Production TTS providers require speaker consent verification before training custom voices. Legal in EU/US.

Audio watermarking AudioSeal (Meta), SynthID (Google) embed imperceptible watermarks in AI-generated audio for provenance.

Deepfake detection DeepFake-o-meter, ADD Challenge 2023 — forensic tools to detect synthetic speech in the wild.

Model cards Responsible TTS systems publish training data sources, bias evaluations, and limitations in model cards.

Regulatory landscape: EU AI Act (2024) classifies real-time voice cloning without consent as high-risk. Transparent labelling of synthetic audio is becoming legally required in multiple jurisdictions.

Key Takeaways

The six essential points from this concept

TTS has four core challenges Homograph ambiguity, prosody inference, naturalness gap, and the vocoder problem. Each generation of TTS models has progressively addressed these.

Gen 1 (Tacotron 2): natural but slow Attention seq2seq → 80-bin Mel → WaveNet vocoder. MOS 4.53. WaveNet makes it 1,000× too slow for real-time — unusable without WaveRNN/HiFi-GAN replacement.

Gen 2 (FastSpeech 2): fast and controllable Parallel non-autoregressive Transformer + HiFi-GAN GAN vocoder. 10× real-time on CPU. Explicit duration/F0/energy control. Industry standard for deployed TTS.

Gen 3 (VALL-E): TTS as language modelling EnCodec tokens replace Mel spectrograms. LLM conditions on 3-sec audio prompt → zero-shot cloning. Paradigm shift equivalent to GPT-3 for text.

HiFi-GAN solves the vocoder problem Multi-period + multi-scale GAN learns to recover phase and synthesise waveforms from Mel spectrograms in real-time. MOS 4.4/5. The current standard open-source vocoder.

Note: Next: AI-MM-IN-TH-000010 — Multimodal Fusion Strategies (early/late/intermediate fusion, cross-modal attention)

AI-MM-IN-TH-000009 Complete

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